moved through space, and that space was a great empty container, a large box which enclosed the universe, an immense structure through which all objects ran true until a force obliged their trajectory to curve. What this ‘space’ was made of, this container of the world he invented, Newton could not say. But a few years before the birth of Einstein two great British physicists, Michael Faraday and James Maxwell, had added a key ingredient to Newton’s cold world: the electromagnetic field. This field is a real entity which, diffused everywhere, carries radio waves, fills space, can vibrate and oscillate like the surface of a lake, and ‘transports’ the electrical force. Since his youth Einstein had been fascinated by this electromagnetic field which turned the rotors in the power stations built by his father, and he soon came to understand that gravity, like electricity, must be conveyed by a field as well: a ‘gravitational field’ analogous to the ‘electrical field’ must exist. He aimed at understanding how this ‘gravitational field’ worked, and how it could be described with equations.

And it is at this point that an extraordinary idea occurred to him, a stroke of pure genius: the gravitational field is not diffused through space; the gravitational field is that space itself. This is the idea of the theory of general relativity. Newton’s ‘space’, through which things move, and the ‘gravitational field’ are one and the same thing.

It’s a moment of enlightenment. A momentous simplification of the world: space is no longer something distinct from matter, it is one of the ‘material’ components of the world. An entity that undulates, flexes, curves, twists. We are not contained within an invisible rigid infrastructure: we are immersed in a gigantic flexible snail-shell. The sun bends space around itself and the Earth does not turn around it because of a mysterious force but because it is racing directly in a space which inclines, like a marble that rolls in a funnel. There are no mysterious forces generated at the centre of the funnel; it is the curved nature of the walls which causes the marble to roll. Planets circle around the sun, and things fall, because space curves.

How can we describe this curvature of space? The most outstanding mathematician of the nineteenth century, Carl Friedrich Gauss, the so-called ‘prince of mathematicians’, had written mathematical formulae to describe two-dimensional undulating surfaces, such as the surfaces of hills. Then he had asked a gifted student of his to generalize the theory to encompass spaces in three or more dimensions. The student in question, Bernhard Riemann, had produced an impressive doctoral thesis of the kind that seems completely